

Hei Chun Sze

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PROFILE

A passionate MSc Computer Game Engineering student at Newcastle University with solid academic achievements. Proficient in C#, C++ and Python. Eager to pursue a career on video game development.

EDUCATION

MSc Computer Game Engineering | Newcastle University Oct 2024 – Current

- Familiarised myself with advanced programming methods in games using C++ in NCLGL, an OpenGL-based renderer provided by the university, in various modules (including a group project).
- Picked up and made use of rendering APIs and techniques to render a scene involving texture mapping, cube mapping and real-time lighting (<https://youtu.be/TpFo-jXn6jM>).
- Built a game by implementing methods focusing on AI character systems and involving calculating physics interactions between objects, so that differently shaped objects respond in a convincing manner (<https://youtu.be/JNAspZGCHcs>).
- Current dissertation on an educational rhythm game built in Unity. Overcame problems like syncing audio clips with a running music track and instantiating different music notes using data-driven charts.

BSc Computer Game Engineering | Newcastle University Oct 2021 – Jun 2024

- Achieved **Upper Second-Class Honours**.
- Used Unity to create a platforming game with a grapple gun for coursework, featuring scene management, AI enemies and physics within a 3D environment.
- Studied multiple algorithms for dissertation on procedural map generation; concluded on which of the picked algorithms is most efficient for in-game application.

A-Levels | Norwich School Sep 2019 – Jun 2021

Computer Science (A), First Language Chinese (A), Mathematics (B), Biology (B)

EXPERIENCE

Teaching Assistant / Substitute Teacher | North East Chinese Association Feb 2025 – Current

- Taught Cantonese to children aged 7 – 10, adapting the teaching pace based on the students' abilities.
- Assigned regular homework and tests, monitoring progress and keeping the children on track for learning.
- Organised conversations in Cantonese for the class; fostered an engaging and supportive environment.

NUGDS Winter Jam 2025 Jan 2025

- Developed a game (<https://itch.io/jam/game-dev-winter-jam-25/rate/3256384>) in Unreal Engine with a newly formed team. Utilised GitHub for version control and managed a limited amount of time.
- Designed procedurally generated rooms using provided assets and blueprints.

VOLUNTEERING

Student Mentor | Norwich Opportunity Area Dec 2019 – Mar 2020

- Mentored and guided younger students to create a fundraiser for a charity of their choice.
- Assisted in discussions and plannings. Ensured the project was delivered within a tight time frame and within the allocated budget; monitored students' financial controllability.

HOBBIES & INTERESTS

Volleyball: Won BUCS National Trophy 2024 as a member of Newcastle University Men's Volleyball 1st Team, awarded Team Royals. Worked constantly to refine hitting form and defence; decision-maker in-game to ensure team morale is maintained.

Blender: Created 3D models of game characters, incorporating provided modelling tools and add-ons to ensure clean topology and structured geometry.

Music: Obtained ABRSM Grade 5 Flute with merit at Norwich School.